



SRv6

Unified solution for AI Networking, SP Mobility and More.

Clarence Filsfils

A 10-year Journey

- 2016 (June): we find SRv6 uSID
 - End-to-End IP solution which empowers the application to control the network
- 2016 (Sept): first week after Leaba acquisition (Silicon One)
 - Confirm S1 Linerate for all required SRv6 use-cases
- 2019: we deploy our first two SRv6 networks (Softbank, Iliad)
 - We are still the only vendor supporting SRv6 on Jericho1 linerate
- 2021: we deploy our first SRv6 uSID's networks (Rakuten, Swisscom...)
- 2023: we deploy Alibaba eCore (DCI/WAN at China scale)
 - A fundamental reference for ongoing AI DCI/WAN deployments
- 2026: within 10 years, we have revolutionized IP and achieved all our deployment targets:
 - SRv6 powers the world's largest AI datacenters (OpenAI, Microsoft, Oracle)
 - More than 100k SRv6 routers deployed across
 - All network domains (DC, DCI, WAN). All markets (Web, SP, Utilities, Ent)
 - Led 13 Proposed Standard RFC's
 - Led open-source implementation in Linux, VPP and SONiC!



SRv6 Unified Solution for AI Networking



Microsoft

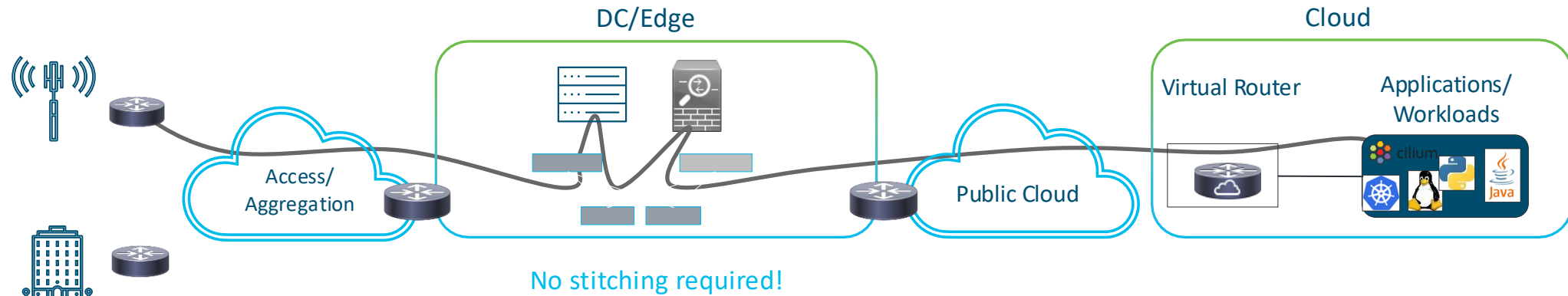
ORACLE

Alibaba

Meta

NEBIUS

SRv6 Unified Solution for SP Mobility



Simplified VPN services

Service Function Chaining

Traffic Engineering and FRR

Operational Simplicity

Enhanced Scalability



Luay Jalil
verizon



Nicolai Leymann
T Deutsche
Telekom



Akash Agrawal
Rakuten

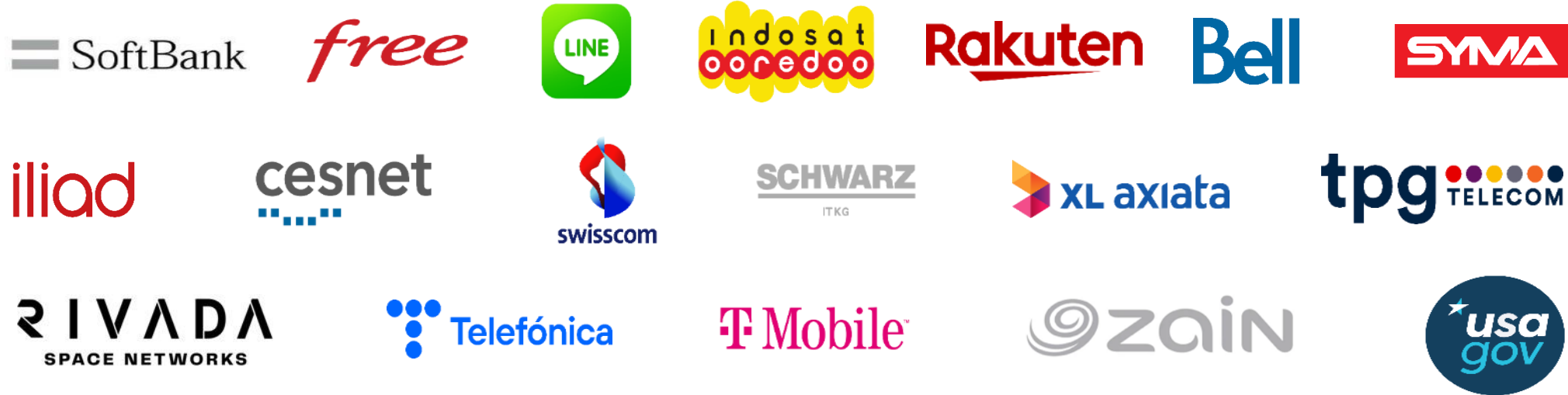


Rolf Schmid

swisscom

Upperside conference Paris

Extensive SP/Utilities Deployments



- Over 100k Cisco SRv6 routers deployed from 2019 to 2026

SRv6 Rich Ecosystem

Network Equipment Manufacturers



Merchant Silicon



Access: A100 Edge: K100
Core: Q200, P100
ToR/Spine: G100, G200



DNX – Jericho
XGS – Tomahawk



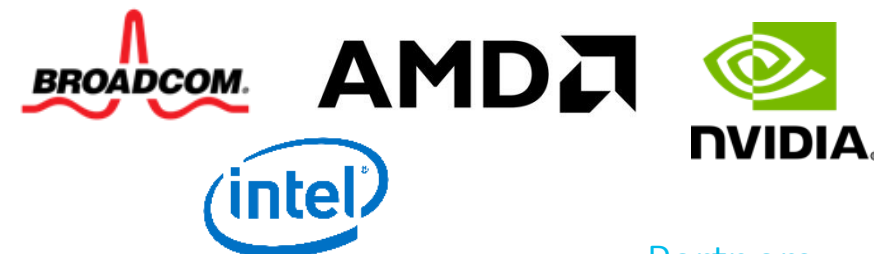
Open-Source Applications



Open-Source Networking Stacks



Smart NIC / DPU



Partners



SRv6 fully Standard at IETF

Architecture

- SR Architecture – **RFC 8402**
- SRTE Policy Architecture – **RFC 9256**
- Compressed SRv6 Segment List – **RFC 9800**

Data plane

- SRv6 Network Programming – **RFC 8986**
- IPv6 SR Header – **RFC 8754**

Control plane

- SRv6 BGP Services – **RFC 9252**
- SRv6 ISIS – **RFC 9352**
- SR Flex-Algo – **RFC 9350**

Operation & management

- SRv6 OAM – **RFC 9259**
- Performance Measurement – **RFC 9503**

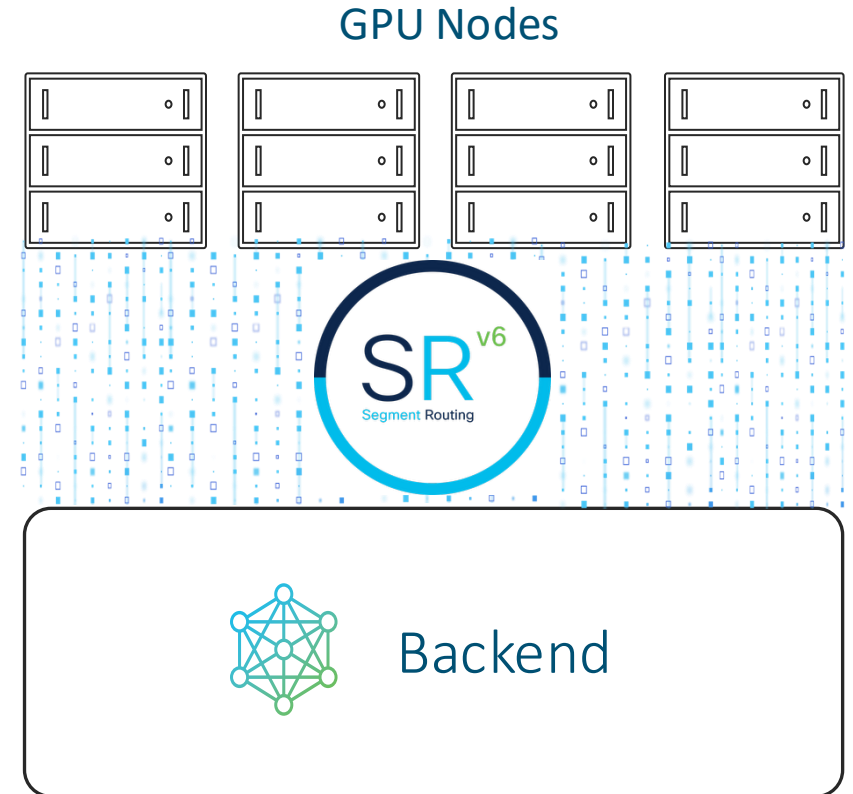
Strong Cisco commitment and leadership

Editor of 96% IETF RFCs
Co-author of 100% IETF RFCs

SRv6 for AI Backend

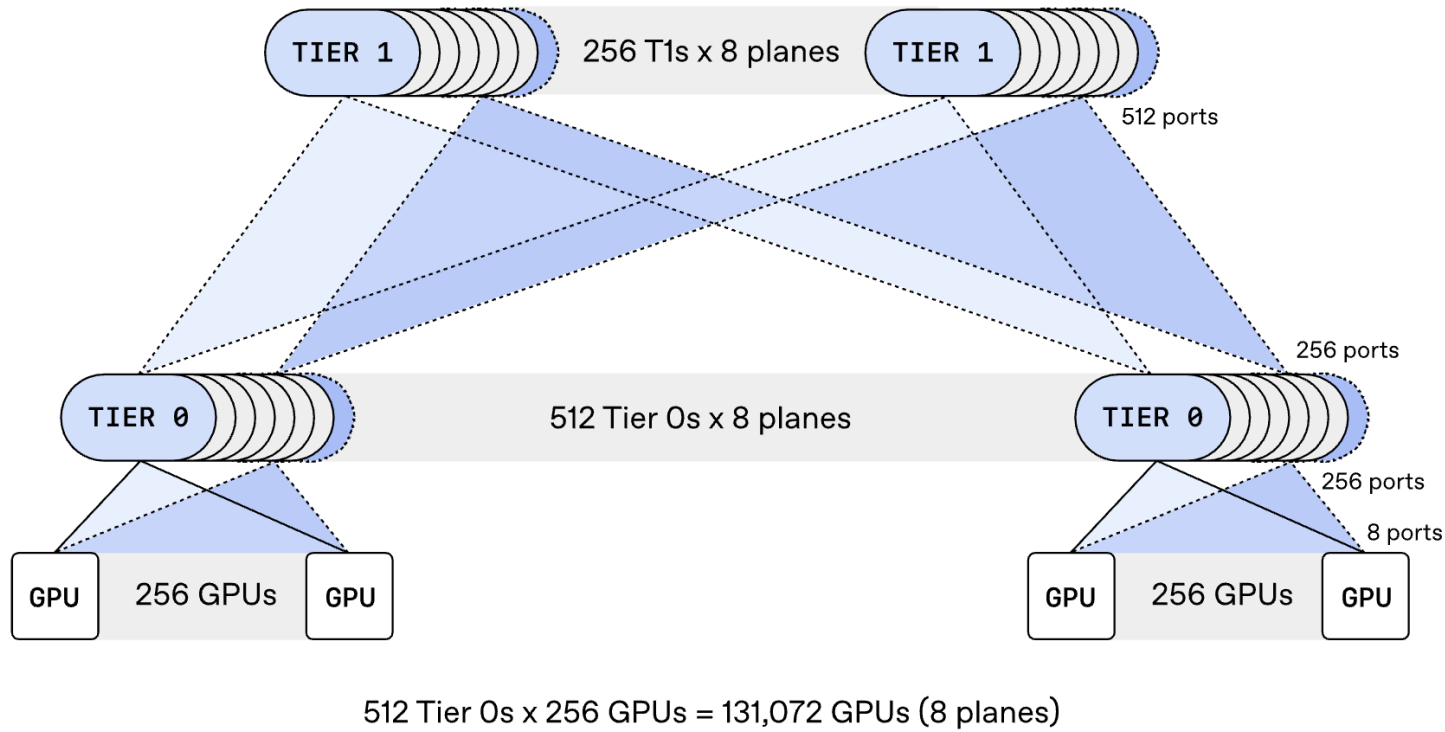
SRv6 for AI Backend

- GPU fabric optimization through **Packet Spraying** over **SRv6 disjoint paths**
- NIC-level source routing via SRv6 uSID
- Stateless, predictable distribution
- Reliable Fabric (no dynamic routing)
- uSec Reaction to congestion/failure
- Deterministic Visibility and Integrated Performance Measurement
 - Path pinning and deterministic feedback



[Clarence Filsfils, Cisco blog: MRC and SRv6 \[Link\]](#).

SRv6 — Powering OpenAI Supercomputer



- Resilient AI Supercomputer Networking using MRC and SRv6 [\[Link\]](#)
- MRC uses SRv6 to spray packets across all network planes, as well as many paths on each plane simultaneously.
- SRv6 allows the sender to encode the full path each packet should take to the network.



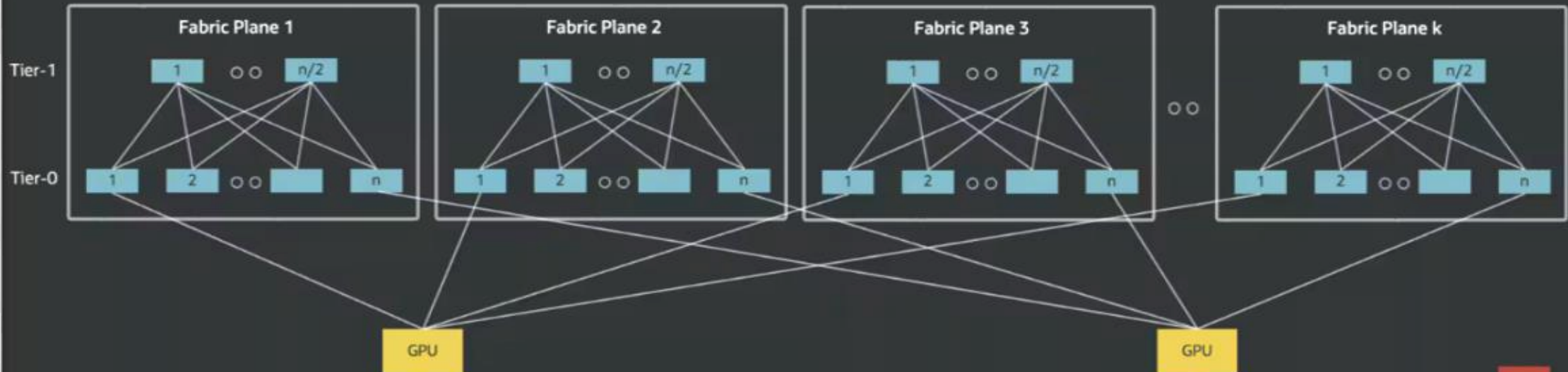
SRv6 — Powering Microsoft's Fairwater AI DC



- Inside the world's most powerful AI datacenter. Microsoft Blog [[Link](#)]
- SRv6 is the key enabling technology for Microsoft DC. Microsoft @ OCP Global Summit [[Link](#)]

SRv6 — Powering Oracle's Stargate AI DC

High Radix: Multiplanar Network Fabric

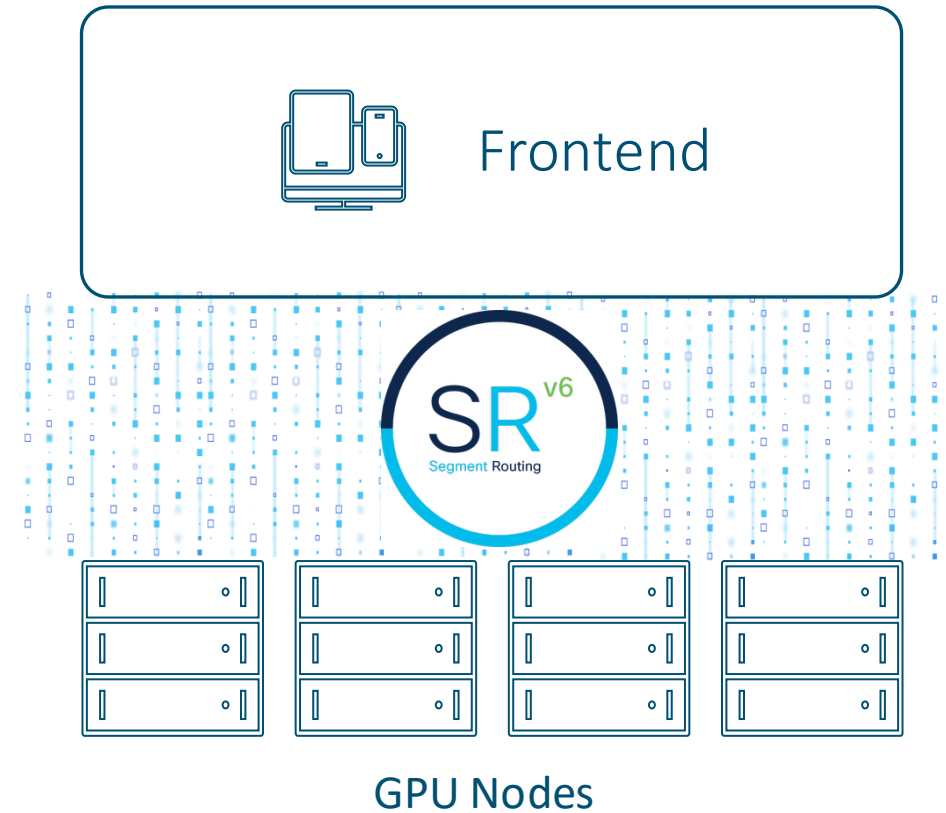


- Oracle Acceleron Multiplanar Networking Architecture [[Link](#)]
- MRC uses SRv6 based static routing. By relying on simple static routing in the network and shifting intelligence to the endpoints, MRC achieves high throughput, resilience, and low tail latency at scale.

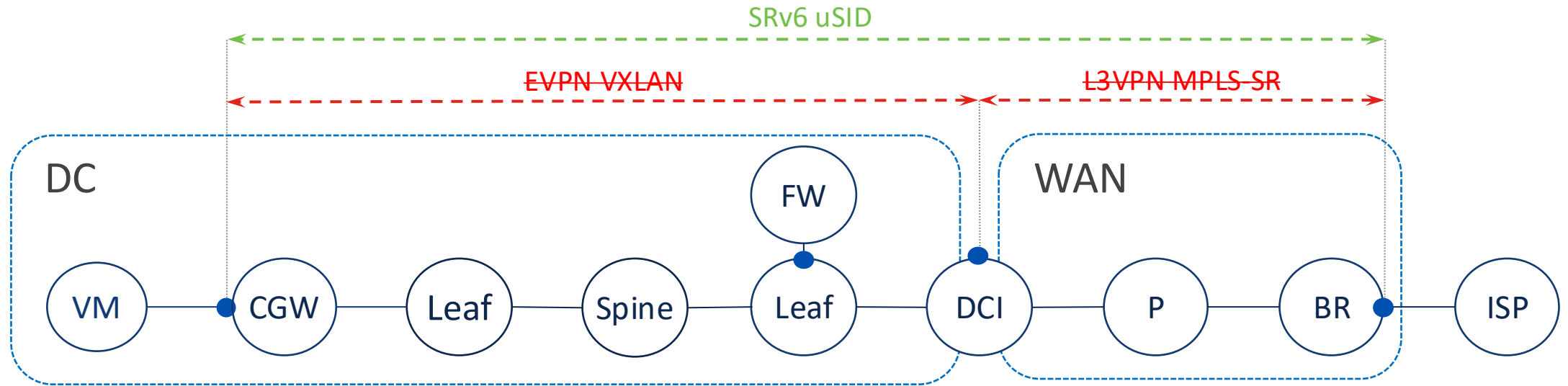
SRv6 for AI Frontend

SRv6 for AI Frontend

- End-to-End unified solution from DC to WAN
 - Elimination of stitching gateways
- Stateless Service Chaining
 - uSID bound to Firewall
- High Availability
 - Anycast uSID bound to Firewall
- Automated Steering
 - Traffic automated steered through service chains to the Internet and from the Internet
- WAN TE controlled from FE
- Egress Peer Engineering controlled from FE



SRv6 - Nebius AI Frontend and WAN

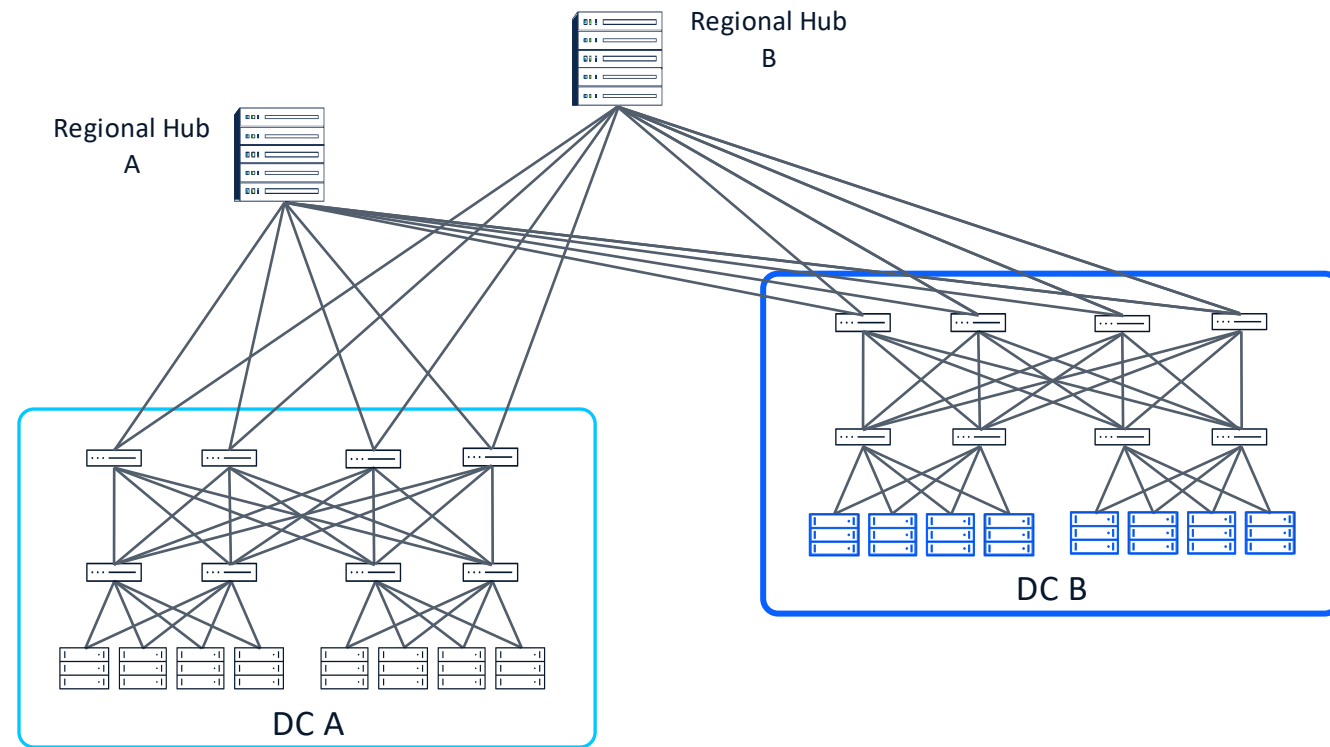


- Unified End-to-End design from the DC Front End, through the metro, up to the Peering with SRv6 (and Firewall Service insertion). Replaces the legacy design with VXLAN in the DC and SR-MPLS in the metro network.
- Alexey Gorovoy, Nebius – Upperside conference, Paris [[Link](#)]

SRv6 for DCI/WAN

DCI/WAN

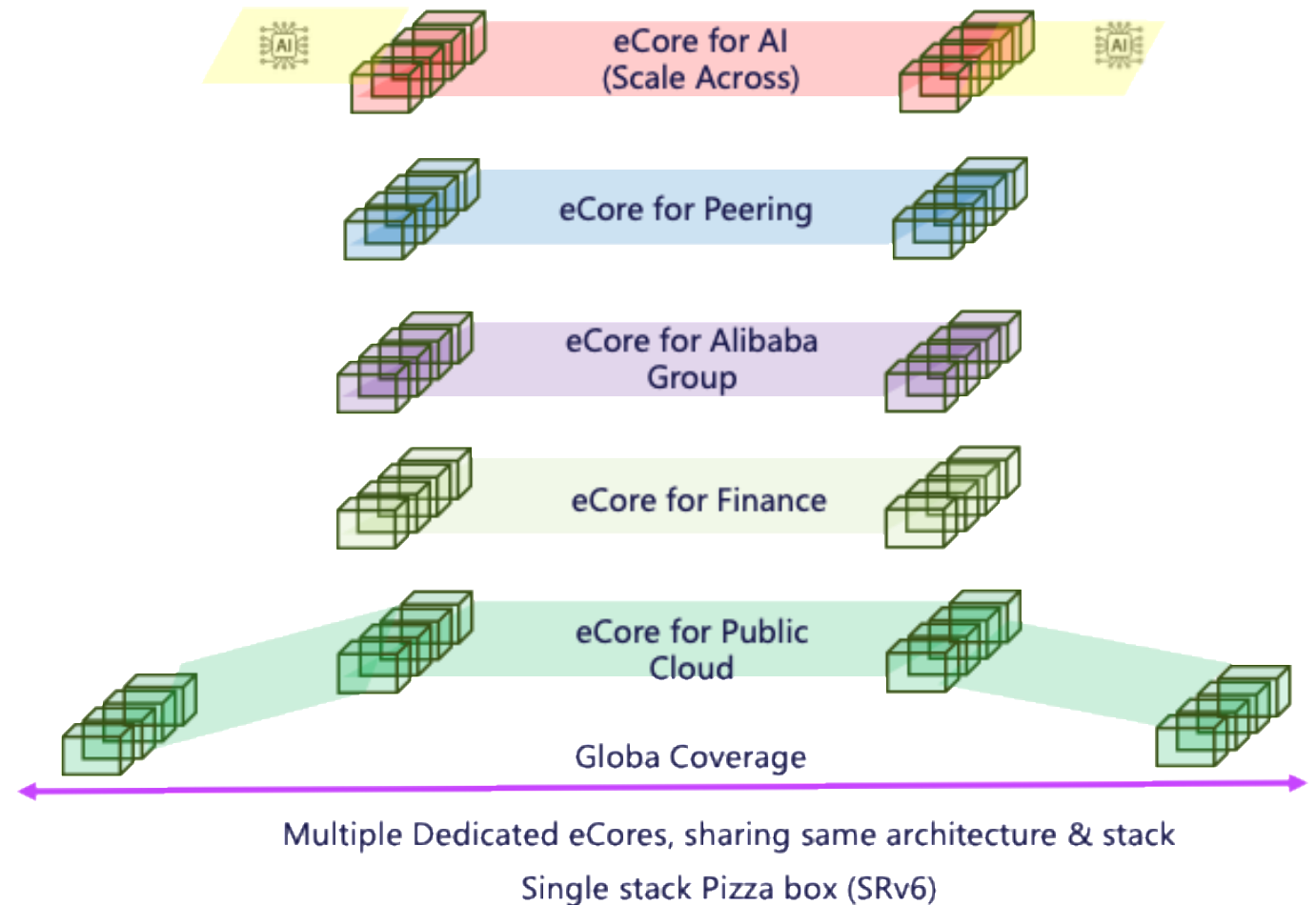
- Power limitations
- Workload spans clusters scattered over a regional network
- Traffic Engineering is MUST
 - Topology is less symmetric
 - Capacity is scarce
 - Latency matters
- SRv6 TE - Perfect fit
 - IP Nature of DCI
 - SONiC expansion in DCI
 - SRv6 rich/mature across DC HW
 - End-to-End - DC to DC through DCI



Clarence Filsfils, Guohan Lu, OCP EMEA 2026 [Link]

Alibaba eCore

- Ultra Scale SRv6 DCI/WAN deployment across 1000 cities in China
- S1/SONiC/SRv6 in DCI
- S1/XR/SRv6 in WAN
- 3 Years of Deployment experience
- Amazing design!
- OCP San Jose (Oct 2026) [[LINK](#)]



Meta – SRv6 Powered Backbone



- SRv6 powers Meta's Regional BackBone (RBB)
- S1/FBOSS/SRv6
- OCP San Jose (Oct 2026) [[Link](#)]

Thu, October 15, 10:30am - 10:45am | SJCC - Concourse Level - 210AE

Srv6 powered backbone networks

● Networking

Backbone networks play a crucial role in connecting Meta's disparate regions. Meta's existing [express backbone network](#) has done this job with aplomb. However, as the number of regions grows, providing this connectivity within a latency envelope becomes much harder. This means we end up replicating core-services infra (which need to provide latency guarantees) in each DC region. This is prohibitive at our scale.

Enter RBB (Regional Back Bone) networks. This is a newly designed inter-region connectivity network running in parallel to our EBB network. Similar to EBB, it's a traffic engineered network. In this session we will go over

- The problems RBB solves.
- Novel use of shallow-buffer switches for long-haul backbone ne
- Why we choose SRv6 as a vehicle for TE in RBB
- How SRv6 is used in this network.

Finally we talk about the SAI support for SRv6 and our experience with it on FBOSS OCP switches using SAI

Speakers



Jasmeet Bagga
Software Engineer - Facebook

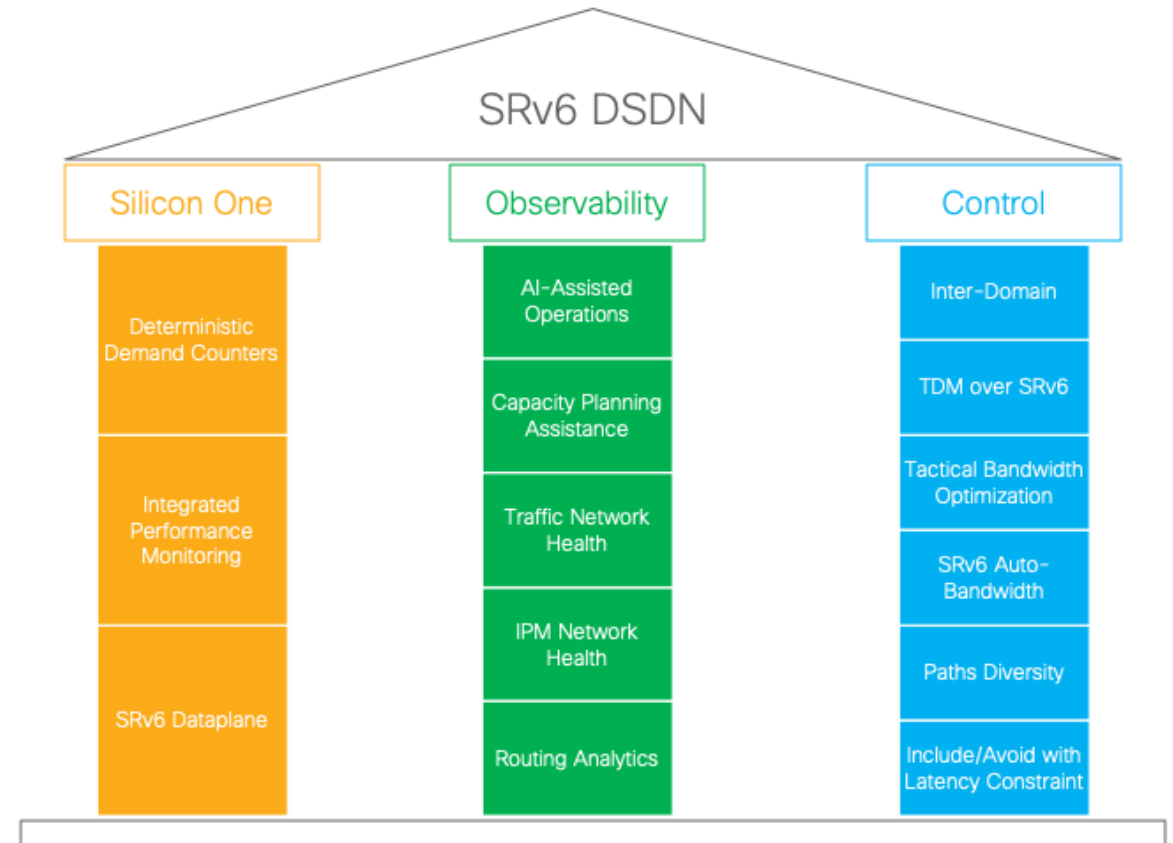


Andrey Golvanov
Network Engineer - Meta

What's Next?

SRv6 DSDN

- Silicon One **enables** everything
- Observability **informs** decisions
- Control **acts** on that intelligence



Three-Pillar Architecture

SRv6 + IPSec



- SRv6 for Path Control
- IPSec for Path Protection
- S1 Native Support:
 - SRv6 Over IPSec
 - IPSec over SRv6
 - SRv6 and IPSec integrated



SRv6 for 6G



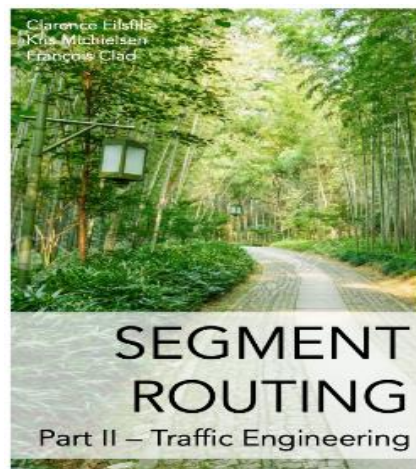
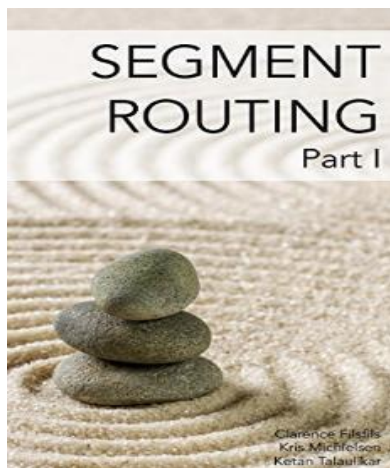
- Great progress at 3GPP!
- 4 Proposals at **3GPP Core Network Protocols WG**
 - SRv6 Baseline Architecture
 - SRv6 Multi-hop forwarding Architecture
 - SRv6 Roaming Architecture
 - Interworking Architecture
- Cosigned by DT, Verizon, Vodafone, Sony, Apple
- Next Meeting Oct 12-16 Prague
 - Operators support is key!

What's Next Next?

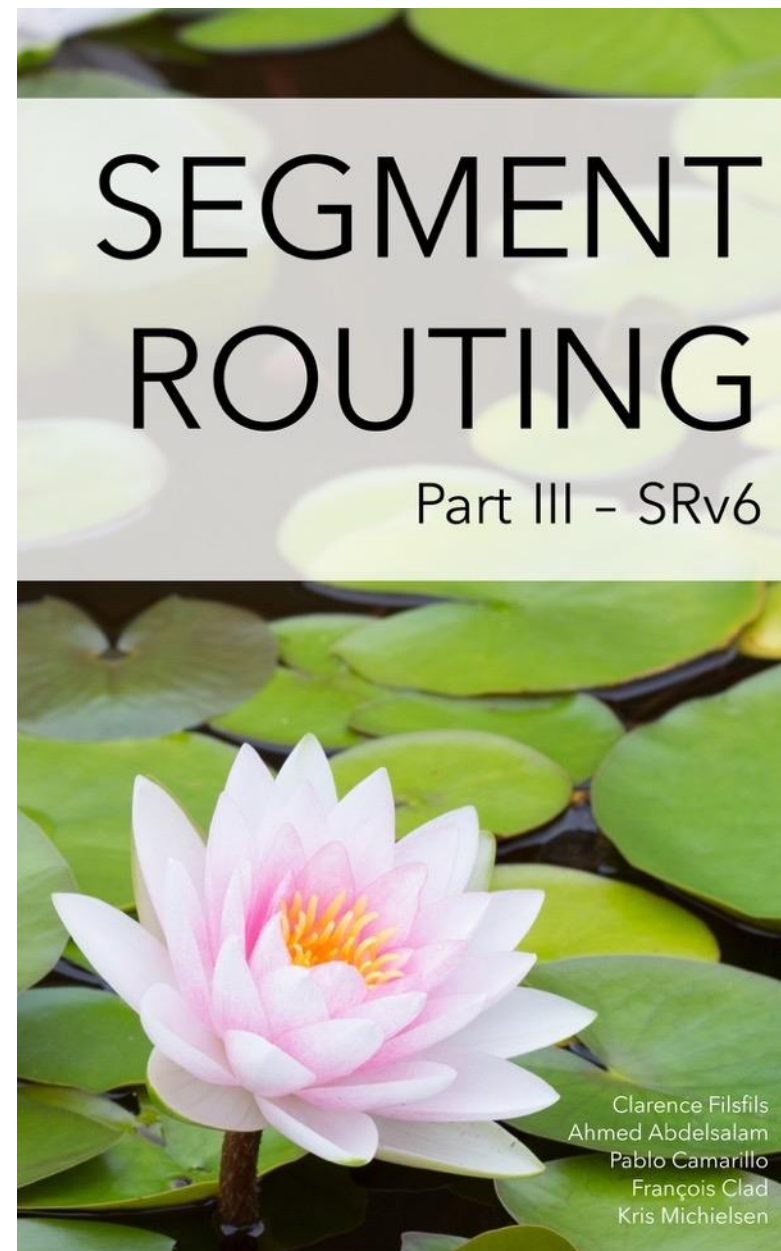
Conclusion

- SRv6 makes IP more scalable, richer with backward compatibility
- SRv6 provides a **Unified** IP Solution
 - AI Fabric: Backend, Frontend, DCI and WAN
 - SP Mobility: from access to Internet through DCI and WAN
- Cisco SR team invented, standardized and deployed across all domains
- Are we done? Not at all. Lots of new SRv6 use-cases!

Stay up-to-date



segment-routing.net



<https://www.amazon.com/dp/B0D6GWWRWH>

